Alvin Kalaivannan

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Education

University of Toronto **Sep. 2026 – Apr. 2030**
Bachelor of Applied Science (B.A.Sc.), Computer Engineering *Toronto, ON*

Experience

Crackd (Scholars Beacon LLC) **Jun. 2025 – Aug. 2025**
Software Engineering Intern *Remote*

- Built internal tools for a SAT/ACT prep platform serving 30K+ students alongside a team of 8 interns and founders.
- Shipped features end-to-end in TypeScript using REST APIs, and wrote automated tests to keep the shared codebase clean.
- Participated in code reviews and picked up tasks directly from the backlog in an agile sprint cycle.

Projects

Apiary AI | *Next.js, TypeScript, Tailwind CSS, OpenTelemetry* **2026**

- Building an AI agent governance and containment tool that lets developers safely run multiple coding agents at once.
- Architecting a five-stage pipeline: OpenTelemetry-based ingestion, a graph-based interaction model, anomaly detection, and git-native containment scoped to developer artifacts.
- Ran a competitive landscape analysis against direct and adjacent players to sharpen product positioning ahead of MVP build.

Overstride | *Python, PyTorch, AthleticsPose, AthletePose3D, Google Colab* **2026**

- Built a two-task injury-classification pipeline from monocular video: population-level screening failed against a demographics control (0/60 pre-registered tests), while within-subject limb identification reached AUC 0.610 on mocap data, with all confound checks verified at chance.
- Measured the real cost of camera-based kinematics against motion capture, finding real video-derived error degrades the limb-identification task from AUC 0.610 to 0.583 while pose estimation itself holds to 3.4° mean joint-angle error.
- Ran six phases of diagnostics (dose-response, learning curves, alternate feature families, a positive control) to confirm the performance ceiling was physiological rather than a modeling limitation, and withheld a per-user demo after finding verdicts flipped on a third of repeat scans.

BananaMOV | *Next.js, TypeScript, Gemini 2.5 Flash, ElevenLabs* **2026**

- Built a VLM-based video analysis pipeline with Gemini 2.5 Flash to extract mood, pacing, and energy from an uploaded video, then passed that analysis to ElevenLabs to generate a custom, royalty-free score.
- Structured the ElevenLabs Music request as a composition plan tied to the video's timeline, so the score evolves with the footage instead of looping underneath it.
- Built stem separation with Demucs and a Web Audio API DAW with mixer, effects, and WAV export; stored projects in Neon Postgres with Vercel Blob for file storage.

VSCS (Vehicle Scanning Collision System) | *Python, OpenCV, PyTorch* **2026**

- Building a computer vision system that scans a vehicle and segments it into individual mechanical components, rather than treating the vehicle as a single bounding box, using footage of a personal van and truck as the scan subject.
- Tracking surrounding objects in the scene frame by frame to estimate collision risk both for the vehicle as a whole and independently per component, so risk can be localized to the specific part nearest a hazard.

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, HTML/CSS, SQL
Frameworks: Next.js, React, Tailwind CSS
Libraries: OpenCV, MediaPipe, PyTorch, AthleticsPose, AthletePose3D, Demucs, Drizzle ORM, TanStack Query, shadcn/ui
Developer Tools: Git, GitHub, Vercel, Clerk, Arduino, ffmpeg, Google Cloud Platform, OpenTelemetry, Neon Postgres, Google Colab
APIs: ElevenLabs, Google Gemini, Google Maps SDK, Web Audio API